

Raw Vibration
(XJTU / FEMTO)

Health Indicator
RMS $\rightarrow$ Log $\rightarrow$ Monotone

Conditional Diffusion
Lookback $\rightarrow$ Forecast

Autoregressive Rollout
 K Trajectories

First-Passage of D
 $\Rightarrow$ RUL Distribution

Rollout-Free Path

Direct-RUL Diffusion (Control)
Window $\rightarrow$ RUL, no Rollout / FPT

Calibration Audit
PIT / Coverage / CRPS